Unit: Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ____________

Date: ____________

Foundation (Jalapeno)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{1}{4\sqrt{2}}$

Q2. Simplify $\frac{1}{\sqrt{3}}$ by rationalizing the denominator.

(A) $\frac{\sqrt{3}}{3}$
(B) $\frac{1}{3\sqrt{3}}$
(C) $\frac{3}{\sqrt{3}}$
(D) $\frac{\sqrt{3}}{9}$
(E) $\frac{1}{9\sqrt{3}}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{5}}$.

(A) $\frac{2\sqrt{5}}{5}$
(B) $\frac{2}{5\sqrt{5}}$
(C) $\frac{5}{2\sqrt{5}}$
(D) $\frac{2\sqrt{5}}{25}$
(E) $\frac{1}{25\sqrt{5}}$

Q4. Simplify $\frac{3}{\sqrt{6}}$ by rationalizing the denominator.

(A) $\frac{3\sqrt{6}}{6}$
(B) $\frac{3}{6\sqrt{6}}$
(C) $\frac{\sqrt{6}}{2}$
(D) $\frac{3\sqrt{6}}{36}$
(E) $\frac{1}{12\sqrt{6}}$

Q5. Rationalize the denominator of $\frac{2}{\sqrt{8}}$.

(A) $\frac{2\sqrt{2}}{4}$
(B) $\frac{2}{4\sqrt{2}}$
(C) $\frac{\sqrt{2}}{2}$
(D) $\frac{2\sqrt{2}}{16}$
(E) $\frac{1}{8\sqrt{2}}$

Q6. Simplify $\frac{5}{\sqrt{10}}$ by rationalizing the denominator.

(A) $\frac{5\sqrt{10}}{10}$
(B) $\frac{5}{10\sqrt{10}}$
(C) $\frac{\sqrt{10}}{2}$
(D) $\frac{5\sqrt{10}}{100}$
(E) $\frac{1}{20\sqrt{10}}$

Q7. Rationalize the denominator of $\frac{4}{\sqrt{12}}$.

(A) $\frac{4\sqrt{3}}{6}$
(B) $\frac{4}{6\sqrt{3}}$
(C) $\frac{2\sqrt{3}}{3}$
(D) $\frac{4\sqrt{3}}{36}$
(E) $\frac{1}{9\sqrt{3}}$

Q8. Simplify $\frac{6}{\sqrt{15}}$ by rationalizing the denominator.

(A) $\frac{6\sqrt{15}}{15}$
(B) $\frac{6}{15\sqrt{15}}$
(C) $\frac{2\sqrt{15}}{5}$
(D) $\frac{6\sqrt{15}}{225}$
(E) $\frac{1}{25\sqrt{15}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{3x}{\sqrt{2x+1}}$ and simplify.

Q10. Rationalize the denominator of $\frac{2y}{\sqrt{y^2-4}}$ and simplify.

Chef's Tips

- Remind students to multiply by the conjugate to rationalize the denominator.
- Encourage students to simplify their answers after rationalizing the denominator.
- Point out that rationalizing the denominator is essential in many mathematical operations.